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Bayesian and conditional frequentist analyses of the Fieller's problem. A critical review

Summary - Fieller's problem belongs to a certain class of problems where there are no methods of "objective" inference that are viewed as broadly satisfactory (Barbieri *et al.* 2001; Berger *et al.*, 1999). In this paper we analyse the problem from different viewpoints: frequentist, likelihood and Bayesian. The usual frequentist solution (Fieller, 1954) is considered from a conditional frequentist viewpoint and it is shown that Fieller's set does not have good conditional properties. We also show that a likelihood analysis fails to be operative. On the other hand, a noninformative Bayesian approach is not well "calibrated" from the frequentist coverage viewpoint. In this paper we argue that it is impossible to obtain a solution to the Fieller's problem which can be considered satisfactory both from frequentist and conditional viewpoints. Namely, we show that the conditional behaviour of the Fieller's set can be very poor, for some value of an appropriate conditioning statistic (or set). We adopt a robust Bayesian approach to show that it is nearly impossible to end up with a reasonable solution to Fieller's problem without introducing some prior information on the parameters.

Key Words - Fieller's Problem; Conditional frequentist inference; Likelihood; Noninformative Bayesian analysis; Robust Bayesian approach.

1. INTRODUCTION

Fieller's problem (Fieller, 1954) is a challenging open question in modern statistical inference. In its simplest form, which is however enough to show the difficulties, the problem can be stated as follows: let x_1 and x_2 be two sample means obtained from n replications of a bivariate normal random variable with mean $\omega = (\omega_1, \omega_2)$ and covariance matrix $\sigma^2 I_2$, where I_2 denotes the 2 by 2 identity matrix. Although the real contexts where Fieller's problem emerges are usually more complicated, this simple setting is enough to show the peculiarities of the problem and to motivate the foundational interest on it. The Fieller's problem amounts to construct a confidence region for $\theta = \omega_1/\omega_2$, that is

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the ratio of the means. For mathematical convenience we will adopt, as the complementary nuisance parameter, either $\lambda = \omega_2$, the mean of X_2, or $\xi = \omega_1^2 + \omega_2^2$. The choice ξ has the attractive feature that ξ is orthogonal to θ in terms of the expected Fisher information matrix. However, the mapping $(\omega_1, \omega_2) \rightarrow (\theta, \xi)$ is not one-to-one and some attention should be paid in using ξ.

The relevance of Fieller's problem lies in the fact that no inferential school is able to provide a completely satisfactory non subjective solution to the problem. From a frequentist point of view, Fieller (1954) proposed to construct a confidence set of level $\gamma = 1 - \alpha$, $\alpha > 0$, based on the pivotal quantity

$$\frac{\sqrt{n}(X_1 - \theta X_2)}{\sigma\sqrt{1 + \theta^2}} \sim N(0, 1) \tag{1}$$

However, this procedure provides regions of infinite size, with positive sampling probability, for each value of θ and for each positive confidence level α.

This result is not surprising because Fieller's problem falls in a category of problems which we will refer to as the GH-class, (Gleser and Hwang, 1987), and which we will illustrate in Section 2. For a GH-problem, no matter what the confidence level α is, the probability of having an unbounded confidence set is greater than zero. This result also implies that Fieller's solution is not a particularly bad frequentist approach to the problem. Basically, there are no frequentist procedures with totally satisfactory properties. Plante (1991) reaches the same conclusion using the idea of inclusion consistency of confidence belts. The Gleser and Hwang result can be used to also show that a conditional analysis based on profile likelihood or some modifications thereof will fail for the same reasons (see Section 2 below or Berger *et al.* (1999)).

A noninformative Bayesian analysis meets problems here. Although the Jeffreys and Reference Prior approaches (Berger and Bernardo, 1992) give proper posterior distributions for the parameter of interest θ, these posteriors are not well calibrated from the frequentist coverage point of view. Because of the propriety of the posteriors, the induced high posterior densities credible regions do turn out to be finite, although sometimes they are disconnected. Indeed, the practical results from a non informative prior Bayesian analysis are often quite different to the results from frequentist approaches.

Because of the GH results, there is no point in trying to find a better unconditional frequentist solution for this problem. A possible alternative approach to Fieller's solution is the conditional frequentist approach (Kiefer, 1977; Berger, 1985). We will analyse several different conditional confidence statements for Fieller's set and we will argue that the one based on an ancillary conditioning is somehow superior.

Finally, a robust Bayesian analysis will be used, to show that it is nearly impossible to end up with a reasonable solution to Fieller's problem without introducing (at least, some) prior information about the nuisance parameter.

2. A CRITICAL REVIEW OF THE EXISTING APPROACHES

2.1. Fieller's solution

Using the pivotal quantity (1), for a fixed $\alpha > 0$, we can produce a $1-\alpha$ level confidence region for θ as the set $C_F(x)$ of values which satisfies the following relation

$$(nx_2^2 - \sigma^2 z_{\alpha/2}^2)\theta^2 - 2nx_1x_2\theta + nx_1^2 - \sigma^2 z_{\alpha/2}^2 \leq 0, \tag{2}$$

where $z_{\alpha/2}$ is the $1-\alpha/2$ percentile of a standard normal distribution. Depending on the observed sample means, the resulting confidence region can be

- an interval, if $x_2^2 > \sigma^2 z_{\alpha/2}^2/n$;
- a cofinite set, if $x_2^2 \leq \sigma^2 z_{\alpha/2}^2/n$ and $x_1^2 + x_2^2 > \sigma^2 z_{\alpha/2}^2/n$;
- the entire real line, if $x_1^2 + x_2^2 \leq \sigma^2 z_{\alpha/2}^2/n$.

It is obviously unappealing to state that the confidence level for $C_F(x)$ is γ when, for example, $C_F(x) = \mathrm{R}$. Also note that the probability of this event is given by

$$P_\omega\left(X_1^2 + X_2^2 \leq \frac{\sigma^2 z_{\alpha/2}^2}{n}\right) = \chi_2^2\left(\frac{n}{\sigma^2}(\omega_1^2 + \omega_2^2), z_{\alpha/2}^2\right), \tag{3}$$

where $\chi_\nu^2(\kappa, t)$ is the cdf of a χ^2 random variable with ν degrees of freedom and non centrality parameter κ, computed at t. This means that Fieller's solution only apparently eliminates the dependence on the nuisance parameter $\xi = \omega_1^2 + \omega_2^2$. Its value actually determines the frequentist probability that $C_F(x)$ will be completely uninformative.

The problems with a frequentist approach to Fieller's problem are masterfully explained in Gleser and Hwang (1987). The statistical problem can be defined as one of *local unidentifiability* of the model, in the following sense. Consider the parametrisation (θ, λ): then as λ goes to zero, θ becomes unidentifiable. We report Theorem 1 of Gleser and Hwang, in our notation.

Theorem 1 (Gleser and Hwang, 1987)**.** *Let $\Omega_1 \times \Omega_2$ be the parameter space and $\mu(\omega_1)$ be a scalar function of ω_1. Suppose that there exists a subset $\Omega_1^* \subset \Omega_1$ and a point $\omega_2^* \in \overline{\Omega_2}$ (the closure of Ω_2) such that $\mu(\omega_1)$ has an unbounded range over Ω_1^* and such that for each value $\omega_1 \in \Omega_1^*$ and for any observation x from the sampling model $p(x|\omega_1, \omega_2)$,*

$$\lim_{\omega_2 \to \omega_2^*} p(x|\omega_1, \omega_2) = p(x|\omega_2^*) \tag{4}$$

exists, and it is a sampling distribution independent of ω_1. Then every confidence procedure $C(x)$ for $\mu(\omega_1)$, with positive confidence level $1 - \alpha$, satisfies

$$P_\omega(d(C(X)) = \infty) > 0$$

for all $\omega = (\omega_1, \omega_2)$, where $d(C(X))$ is the diameter of C(X).

In our context, by setting $\mu(\omega_1) = \theta$, $\Omega_1^* = (k, \infty)$ for some constant k, and $\omega_2^* = 0$, one obtains that no confidence procedures with expected finite diameter exist for θ.

2.2. Likelihood Analysis

The Gleser and Hwang result is important also from a likelihood point of view. As a general result (Berger et al, 1999), if condition (4) is satisfied, then the profile likelihood $\hat{L}(\theta)$, obtained by maximisation of the likelihood function with respect to the nuisance parameters, is bounded away from zero. In Fieller's problem the profile likelihood turns out to be

$$\hat{L}(\theta) \propto \exp\left\{-\frac{n(x_1 - \theta x_2)^2}{2\sigma^2(1+\theta^2)}\right\}, \tag{5}$$

and, as $|\theta|$ goes to ∞, the profile likelihood approaches the value $\exp\{-nx_2^2/(2\sigma^2)\}$. For moderate to large values of the sample ratio x_1/x_2, any likelihood set based on $\hat{L}(\theta)$ will be unbounded. See Brown and Sundberg (1987) for related considerations.

One could also consider using a conditional or marginal likelihood based on the marginal or conditional distribution of some components of the sufficient statistics. These approaches also fail, as we now show. From (3) we may note that $T = X_1^2 + X_2^2$ is an ancillary statistic for θ. Let (R, T) be the jointly sufficient statistics for (θ, ξ), where $R = X_1/X_2$. Then the conditional distribution of R, given the θ-ancillary $T = t$, can be shown to be

$$p(r|t, \theta, \xi) = \frac{1}{\pi(1+r^2)} \frac{\cosh\left(\frac{n\sqrt{\xi t}(1+r\theta)}{\sigma^2\sqrt{(1+r^2)(1+\theta^2)}}\right)}{I_0\left(\frac{n\sqrt{\xi t}}{\sigma^2}\right)}, \tag{6}$$

where I_0 is the modified Bessel function of the first kind and of order zero.

Also, the marginal distribution of R can be derived from Marsaglia (1965) to be, in the (θ, λ) parametrisation,

$$p(r|\theta,\lambda)=\frac{\exp\{-n\lambda(1+\theta^2)/(2\sigma^2\}}{\pi(1+r^2)}\left[1+\frac{q(r,\theta,\lambda)}{\varphi(q(r,\theta,\lambda))}\left(\Phi(q(r,\theta,\lambda))-\frac{1}{2}\right)\right], \quad (7)$$

where φ and Φ represent the pdf and the cdf of a standard normal distribution and

$$q(r,\theta,\lambda)=\frac{\lambda(1+r\theta)\sqrt{n}}{\sigma\sqrt{1+r^2}}.$$

Both approaches fail to eliminate the nuisance parameter, either ξ or λ. Also the implied reduced statistical models still suffer from the Gleser and Hwang pathology. In fact

$$\lim_{\xi\to 0} p(r|t,\theta,\xi)=\lim_{\lambda\to 0} p(r|\theta,\lambda)=\frac{1}{\pi(1+r^2)}, \quad (8)$$

a standard Cauchy density, independently of θ.

Even using a noninformative prior to eliminate the nuisance parameter does not solve the problem. It can be easily shown that the conditional reference prior (Sun and Berger, 1998; Berger *et al.*, 1999) for $\xi|\theta$ is

$$\pi^R(\xi|\theta)\propto\frac{1}{\sqrt{\xi}} \quad (9)$$

and the corresponding integrated likelihood

$$\tilde{L}(\theta)=\int_{\Xi} p(x|\theta,\xi)\pi^R(d\xi|\theta)$$

coincides with the profile likelihood.

3. Bayesian noninformative analysis

The noninformative Bayesian approach to the problem only requires the selection of a default prior distribution for the parameters. Then one computes, if it exists, the posterior distribution, and derives the marginal posterior for θ. For Fieller's problem it is shown in Liseo (1993) (see Creasy (1954) for a fiducial interpretation of this prior) that the Jeffreys' prior is

$$\pi^J(\lambda,\theta)\propto|\lambda|, \quad (10)$$

while the reference prior approach (Berger and Bernardo, 1992) gives

$$\pi^R(\lambda, \theta) \propto \frac{1}{\sqrt{1+\theta^2}} \tag{11}$$

[Liseo, 1993; see also Kubokawa and Robert (1994) for more general results]. It is also easy to see that both priors give proper posterior distributions. Standard calculations show that

$$\pi^J(\theta|x_1, x_2) = \frac{1}{\pi(1+\theta^2)} \exp\left\{-\frac{n(x_1^2+x_2^2)}{2\sigma^2}\right\}\left[1+\frac{s(\theta)}{\varphi(s(\theta))}\left(\Phi(s(\theta)) - \frac{1}{2}\right)\right], \tag{12}$$

where

$$s(\theta) = \frac{\sqrt{n}(x_2 + \theta x_1)}{\sigma\sqrt{1+\theta^2}}. \tag{13}$$

The reference marginal posterior distribution is proportional to

$$\pi^R(\theta|x_1, x_2) = \frac{1}{\pi(1+\theta^2)} \exp\left\{-\frac{n(x_1 - \theta x_2)^2}{2\sigma^2(1+\theta^2)}\right\} \frac{\exp\left\{\frac{n(x_1^2+x_2^2)}{4\sigma^2}\right\}}{I_0\left(\frac{n(x_1^2+x_2^2)}{4\sigma^2}\right)}. \tag{14}$$

Both the posteriors have thick (Cauchy-like) tails and, depending on the observed x, they can be either unimodal or bimodal. Using Marsaglia (1965) it can be seen that, for values of $x_2\sqrt{n}/\sigma$ greater than 3, then (12) is unimodal if and only if $x_1\sqrt{n}/\sigma > 2.257$. Although analytically different, the two marginal posteriors are typically indistinguishable graphically. Being proper, both (12) and (14) provide finite, although sometimes disconnected, credible sets. However, credible sets derived from (12) or (14) fail to have good frequentist coverage probability for any fixed sample size n; this happens even if they asymptotically have the "correct" coverage. This apparent contradiction is clarified in Gleser and Hwang (1987) as arising from a technical reason: the limits as $n \to \infty$, and as $\omega_2 \to 0$, cannot be interchanged. Then, for fixed θ, as $n \to \infty$ the coverage of the reference prior is fine; however, for fixed n, there is always a subset of the parameter space where the coverage is poor.

There is also an alternative "non standard" noninformative Bayesian analysis: the geometry of the problem may suggest to work with $\psi = \tan^{-1}\theta$ as the parameter of interest. Then the use of a uniform proper prior over $\Delta = [-\pi/2, \pi/2]$ would correspond to a Cauchy prior in the θ parametrisation. A default conditional prior for the nuisance parameter λ can be derived, for example, following Sun and Berger (1998). They show how to derive a conditional reference prior for ψ_2 given ψ_1 where $\psi = (\psi_1, \psi_2)$ is the parameter vector, and the marginal prior for ψ_1 is assumed to be known. To find the

conditional prior it is necessary to define a nested sequence of compact subsets of the parameter space (see Sun and Berger (1998) for details). In the Fieller's problem it is natural to choose rectangles in the (ω_1, ω_2) parametrisation, so that

$$\Omega_m = \{(\omega_1, \omega_2) : -a_m < \omega_1 < a_m, -b_m < \omega_2 < b_m\} \qquad m = 1, 2, \ldots ,$$

where $\{a_m\}$ are increasing sequences diverging to ∞. Then it is possible to show that the conditional reference prior for λ given θ is

$$\tilde{\pi}(\lambda|\theta) \propto 1. \tag{15}$$

The resulting marginal posterior distribution for θ is proportional to

$$\tilde{\pi}(\theta|x) \propto \frac{1}{(1+\theta^2)^{3/2}} \hat{L}(\theta) = \frac{1}{1+\theta^2} \pi^R(\theta|x). \tag{16}$$

The behaviour of $\tilde{\pi}(\theta|x)$ is qualitatively similar to that of the full reference marginal posterior; however tails are thinner, as is apparent from (16). It makes Bayesian conclusions even more different from those obtained in a frequentist way. This is an obvious consequence of using a proper marginal prior on the parameter of interest θ.

4. Conditional frequentist analysis of Fieller's set

4.1. Background

It is clear that Fieller's set has post-data troubles when the observed means are close to zero, and it seems reasonable that a confidence level report should be in some way dependent on the observed data. The theoretical development of a conditional frequentist theory of inference dates back to two seminal papers by Kiefer (1975, 1977), although these papers could be defined as an effort to formalise ideas of Fisher and Cox. Berger (1985,1988), Brown (1988) and Casella (1988,1992) have given new impetus to this approach, emphasizing the necessity in many problems for at least partial conditioning on the observed data. Goutis and Casella (1995) a good reference paper on these topics.

Here we will concentrate our analysis on confidence procedures. There are basically two separate approaches to the definition of conditional confidence level reports, namely the *estimated confidence level* (ECL) and the *conditional confidence level* (CCL).

In the ECL approach one reports, as a confidence level for a given confidence procedure $C(X)$, a data dependent measure of accuracy $\gamma(x)$. The word

estimated comes from the fact that $\gamma(x)$ can be interpreted as an estimator of the unknown parametric indicator function

$$I_C(\theta) = \begin{cases} 1 & \theta \in C(X) \\ 0 & \theta \notin C(X). \end{cases} \tag{17}$$

A good estimator of $I_C(\theta)$ should satisfy the following frequentist validity constraint (Berger, 1985):

$$\forall \theta \in \Theta, \qquad E_\theta\,(\gamma(X)) \leq P_\theta(C(X)) \tag{18}$$

In words, the frequentist average reported level should not be greater than the nominal value, for each single θ. In the CCL approach, one partitions the sample space as

$$\mathcal{X} = \bigcup_{s \in \mathcal{S}} \mathcal{X}_s, \qquad \mathcal{X}_s \cap \mathcal{X}_r = \emptyset, \quad \textit{for} \;\; r \neq s,$$

and then develops a frequentist inference, conditional on the observed $\mathcal{X}_s$. In a set estimation context, one reports, as a conditional confidence level

$$\gamma(s) = \inf_\theta P_\theta(\theta \in C(X) | \mathcal{X}_s). \tag{19}$$

This approach is very much related to the ideas of Relevant Subsets (Fisher, 1956; Robinson, 1979) and ancillary statistics, which play a central role in the theory of conditional inference. The conditional confidence level automatically satisfies the frequentist validity constraint because

$$E_\theta\,(\gamma(s)) \leq E_\theta\,[P_\theta(\theta \in C(X) | \mathcal{X}_s)] = P_\theta(\theta \in C(X)). \tag{20}$$

The main drawback of the conditional confidence approach is that its performance depends on the actual partition we adopt, and any particular choice is often difficult to justify as superior to another. However, for Fieller's problem, we will see that a natural choice stems from the peculiar behavior of Fieller's set.

4.2. Conditional confidence levels for $C_F(X)$

Tsao and Hwang (1998, 1999) have proposed two different estimated confidence levels for the accuracy of $C_F(X)$. The first one,

$$\gamma_+ = \begin{cases} \gamma & C_F(X) \neq \mathrm{R} \\ 1 & C_F(X) = \mathrm{R}, \end{cases} \tag{21}$$

is a trivial improvement upon the unconditional level γ; when $C_F(X) = \mathrm{R}$ it seems more reasonable to report confidence 1 with the assertion "$\theta \in \mathrm{R}$"!

The second estimated confidence level of Tsao and Hwang (1998, 1999) is

$$\gamma_L(x) = \int_{C_F(X)} \pi_J(d\theta|x), \tag{22}$$

that is, the posterior probability, under the Jeffreys' prior, of Fieller's set. This is obviously different from using the Jeffreys' highest posterior credible region; the region itself is the Fieller's region; but its accuracy is measured in a default Bayesian way. Tsao and Hwang (1998, 1999) show that, under squared error loss,

$$W(\theta, \gamma(x)) = (\gamma(x) - I_{C_F}(\theta))^2, \tag{23}$$

both γ_+ and γ_L dominate γ. Unfortunately both the proposed estimated confidence levels fail to satisfy the frequentist validity condition (18). This is easy to see because, for all $x \in \mathcal{X}$, $\gamma_+ \geq \gamma$, and $\gamma_L \geq \gamma$ (Tsao and Hwang, 1999).

The peculiar behavior of $C_F(X)$, which turns out to be an interval, a cofinite set or the entire real line, depending on the observed value of x, can be suitably used to construct a conditional confidence level for $C_F(X)$. This can be done by simply splitting the sample space, $\mathcal{X}$, into three subsets, namely

- $\mathcal{X}_1 = \{(x_1, x_2) : C_F(x) \textit{is bounded}\}$;
- $\mathcal{X}_2 = \{(x_1, x_2) : C_F(x) \textit{is cofinite}\}$;
- $\mathcal{X}_3 = \{(x_1, x_2) : C_F(x) = \mathrm{R}\}$.

Accordingly, we define a CCL $\gamma_c(x)$ as

$$\gamma_c(x) = \begin{cases} \inf_\omega P_\omega(C_F(X)|\mathcal{X}_1) & x \in \mathcal{X}_1 \\ \inf_\omega P_\omega(C_F(X)|\mathcal{X}_2) & x \in \mathcal{X}_2 \\ 1 & x \in \mathcal{X}_3. \end{cases} \tag{24}$$

The next theorem shows that, by conditioning on the above partition, the confidence level of $C_F(X)$ can be zero. It actually happens for $x \in \mathcal{X}_1$. Also, for $x \in \mathcal{X}_2$, the conditional level will be a function of the original γ.

Theorem 2. *For any* $\gamma = 1 - \alpha > 0$,

$$\gamma_c(x) = \begin{cases} 0 & x \in \mathcal{X}_1 \\ \nu(\alpha) & x \in \mathcal{X}_2 \\ 1 & x \in \mathcal{X}_3, \end{cases} \tag{25}$$

where

$$\nu(\alpha) = \frac{(1-\alpha)^2 - (1 - e^{-z^2_{\alpha/2}})}{(1-\alpha) - (1 - e^{-z^2_{\alpha/2}})}.$$

Proof. See the Appendix. □

For example, when $\alpha = .05$, $\nu(\alpha)$ is approximately .508. This means that the confidence level $\gamma = .95$ for $C_F(X)$ is actually a "compromise" value among three extremely different conditional confidence levels.

Theorem 2 can be viewed as a major criticism of Fieller's solution from a conditional point of view. The unconditional confidence report is, most of the time, at least misleading. However, something can be recovered by considering a coarser partition of $\mathcal{X}$, namely $\mathcal{Y}_1 = \mathcal{X}_1 \bigcup \mathcal{X}_2$ and $\mathcal{Y}_2 = \mathcal{X}_3$. This way we can avoid a zero confidence level. In fact, by defining

$$\gamma_c'(x) = \begin{cases} \inf_\omega P_\omega(C_F(X)|\mathcal{Y}_1) & x \in \mathcal{Y}_1 \\ 1 & x \in \mathcal{Y}_2, \end{cases} \tag{26}$$

the following theorem can be proved.

Theorem 3. *For any* $\gamma = 1 - \alpha > 0$,

$$\gamma_c'(x) = \begin{cases} \nu'(\alpha) & x \in \mathcal{Y}_1 \\ 1 & x \in \mathcal{Y}_2, \end{cases} \tag{27}$$

where

$$\nu'(\alpha) = \frac{e^{-z_{\alpha/2}^2} - \alpha}{e^{-z_{\alpha/2}^2}}.$$

Proof. See the Appendix. □

Note that the quantity $\nu'(\alpha)$ is always greater than zero for $\alpha > 0$ and it tends to zero, as $\alpha \to 1$. It is interesting to evaluate the performance of this estimator in terms of frequentist risk against the squared error loss (23). For $x \in \mathcal{Y}_2$, the conditional risk is obviously zero, while

$$R(\omega, \gamma_c'|\mathcal{Y}_1) = E_\omega\left((\gamma_c') - I_{C_F}(\theta))^2|\mathcal{Y}_1\right) = (\gamma_c')^2 - (2\gamma_c' - 1)\frac{P_\omega(C_F \cap \mathcal{Y}_1)}{P_\omega(\mathcal{Y}_1)}.$$

Since $C_F \cap \mathcal{Y}_2 = \mathcal{Y}_2$ no matter what ω is (when C_F is considered a subset of $\mathcal{X}$), one obtains

$$R(\omega, \gamma_c'|\mathcal{Y}_1) = (\gamma_c')^2 - (2\gamma_c' - 1)\frac{\gamma - P_\omega(\mathcal{Y}_2)}{1 - P_\omega(\mathcal{Y}_2)},$$

where $P_\omega(\mathcal{Y}_2)$ is given by (3). This risk is a function of $\|\omega\|^2 = \omega_1^2 + \omega_2^2$. As $\|\omega\|$ approaches zero, $R(\omega, \gamma_c'|\mathcal{Y}_1)$ will tend to $\gamma_c'(1 - \gamma_c')$. This formula mimics the one for the unconditional risk, which is uniformly $\gamma(1-\gamma)$. Note that the estimator γ_c' is not comparable with the others given in Tsao and Hwang (1998, 1999) because those do not satisfy condition (18). Note also that the conditional risk of γ_c' is uniformly larger than the unconditional one when $x \in \mathcal{Y}_1$, and uniformly smaller when $x \in \mathcal{Y}_2$, as the intuition suggests.

5. Conditional confidence level based on an ancillary statistic

In the previous section we tried to avoid the occurrence of a zero conditional level by using a coarser partition of the sample space. Going in the opposite direction, one could also think of conditioning on a partition which is as fine as possible. Apart from the trivial partition (single points of $\mathcal{X}$), a sensible partition can be obtained by conditioning on the ancillary statistic $T = X_1^2 + X_2^2$. This way it is possible to define another conditional confidence level for C_F, namely

$$\gamma_a(t) = \inf_{\omega} P_\omega(C_F|T = t). \tag{28}$$

Using the conditional distribution of $R|T$ in (6) it is readily seen that

$$P_\omega(C_F|T = t) = P_\omega\left(\frac{(R-\theta)^2}{1+R^2} \leq \frac{\sigma^2(1+\theta^2)z_{\alpha/2}^2}{nT} | \, T = t\right).$$

Let $k = \sigma^2 z_{\alpha/2}^2/n$. When $t < k$,

$$\frac{(R-\theta)^2}{1+R^2} < (1+\theta^2) < \frac{k}{t}(1+\theta^2)$$

for all values of $R = r$ and θ, and $\gamma_a(t) = 1$. When $t \geq k$, the confidence level is given by

$$\gamma_a(t) = \inf_{\omega} \begin{cases} \displaystyle\int_{r_2(\theta,t)}^{r_1(\theta,t)} p(r|t,\theta,\xi)dr & t < k(1+\theta^2) \\ \displaystyle 1 - \int_{r_1(\theta,t)}^{r_2(\theta,t)} p(r|t,\theta,\xi)dr & t \geq k(1+\theta^2) \end{cases}$$

where $r_1(\theta,t)$ and $r_2(\theta,t)$ are given by

$$\frac{\theta t \pm (1+\theta^2)\sqrt{k(t-k)}}{t - k(1+\theta^2)}.$$

Unfortunately, a numerical analysis of the function $\gamma_a(t)$ shows that is always close to zero; then $\gamma_a(t)$ cannot be considered a valid confidence level for C_F. We are currently studying two other possible versions of $\gamma_a(t)$. The first one is an *estimated* version, namely

$$\gamma_a(t, \hat{\xi}) = \inf_{\theta} P_{\theta,\hat{\xi}}(C_F \mid T = t);$$

it amounts to estimating the nuisance parameter ξ and therefore avoiding to compute the infimum over this parameter. The second one is based on the

heuristic that, as $\|\omega\| \to 0$, the conditional distribution of $R \mid T = t$ approaches a standard Cauchy density: this means that we can use, as an approximate conditional confidence level, the quantity

$$\gamma_a(t) = \begin{cases} 1 & t \leq k \\ \dfrac{2}{\pi} \tan^{-1}\left(\sqrt{\dfrac{k}{t-k}}\right) & t > k. \end{cases} \tag{29}$$

The behavior of $\gamma_a(t)$ is very reasonable. It is 1 for all $t < k$, that is, when C_F corresponds to the real line, and then it decreases from 1 to 0 as t increases, and the size of C_F decreases. This is a further indication that Fieller's approach does not have good uniform conditional properties. Note also that $\gamma_a(t)$ satisfies, by definition, the frequentist validity condition. In terms of risk, it can be easily shown that

$$R(\omega, \gamma_a(t)) = \gamma_a^2(t) - (2\gamma_a^2(t) - 1) P_\omega(C_F | T = t).$$

Again, as $\|\omega\| \to 0$, the conditional risk will approach $\gamma_a(t)\,((1 - \gamma_a(t))$.

6. Robust Bayesian Analysis

Another way of analysing the sensitivity of the answers on θ with respect to the nuisance parameter λ is to take a robust Bayesian approach (see Berger (1994) for a comprehensive discussion of the topic) where a class Γ of prior distributions is considered instead of a single one and sensitivity of the answer with respect to the prior is investigated.

In Fieller's problem, for example, we are interested in analysing to what extent values of λ determine the precision of the answers on θ, and one could consider the following class of priors:

$$\Gamma(k, \alpha) = \{\pi : \pi(\theta, \lambda) = \pi_0(\theta)\pi(\lambda) \text{ and } \Pi(|\Lambda| < k) < \alpha\}, \tag{30}$$

where $= \pi_0(\theta)$ is a fixed prior on θ, and Π denotes the probability measure associated with the density π. In the following analysis we will assume $\pi_0(\theta)$ to be a Cauchy(0,1) distribution. There are several ways to justify the use a Cauchy prior for θ. For example, it can be seen as the result of assuming conjugate $N(0, \tau_i^2)$ priors on ω_i, $i = 1, 2$, and then taking large values for the τ_i^2's in such a way that $\tau_1^2/\tau_2^2 \to 1$. As suggested in Section 2.3 it is also equivalent to assume a uniform prior over $\psi = \tan^{-1}\theta$. The class of priors we propose here is a simple extension of the quantile class discussed in Ruggeri (1991, 1993). We consider a weakly informative fixed prior on θ, and only specify ranges of prior probabilities for the Λ-partition $\{\Lambda_k, \Lambda_k^c\}$,

where $\Lambda_k = \{|\Lambda| \geq k\}$. The analysis of these classes over a reasonable range of values of k and α will provide an idea of the sensitivity of the posterior answers with respect to the prior probability of Λ_k. Note also that multidimensional robustness is rarely achieved without assuming strong conditions on the prior, such as independence among the components (Berger, 1994). Our goal is to show that, even with rather restricted classes of priors, the final results will be typically sensitive to the choice of k. Also, when $x_1^2 + x_2^2$ is small, posterior robustness is typically impossible to achieve, and final reports will heavily depend on k and α. The following theorem is easy to prove.

Theorem 4. *Consider, for fixed values of* k $(k > 0)$ *and* α $(0 < \alpha < 1)$, *the class of priors* (30). *Let* B *any measurable subset of* Θ. *Then*

$$\sup_{\pi \in \Gamma(k,\alpha)} P(B|x) = \sup_{|w|<k, |v|\geq k, 0<\xi<\alpha} \frac{\xi \tilde{L}_B(w) + (1-\xi)\tilde{L}_B(v)}{\xi \tilde{L}_\Theta(w) + (1-\xi)\tilde{L}_\Theta(w)}, \tag{31}$$

where

$$\tilde{L}_A(\lambda) = \int_A f(x|\theta, \lambda)\pi_0(d\theta).$$

Replacing "sup" by "inf" yields the analogous result for the infimum.

Proof. Trivial. □

Also note that the right hand expression in (31) is monotone in ξ (decreasing or increasing, depending on the set B). Therefore the optimal value for ξ will be always either 0 or α.

TABLE I

Case I: $x = 2$; $y = 2$					
		Fieller's Set		Jeffreys' Set	
k	α	inf	sup	inf	sup
.5	.1	.05	.17	.06	.22
.5	.5	.05	.17	.06	.22
.5	.9	.05	.07	.06	.22
1	any	.41	.88	.49	.97
2	any	0	.95	0	.98
3	any	0	.75	0	.90
4	any	0	.36	0	.68

TABLE II

Case II: $x = 2$; $y = .2$					
		Fieller's Set		Jeffreys' Set	
k	α	inf	sup	inf	sup
.1	.1	.97	.98	.97	.99
.1	.3	.96	.98	.93	.99
.1	.5	.95	.98	.87	.99
.1	.7	.94	.98	.81	.99
.1	.8	.91	.98	.81	.99
.1	.9	.85	.98	.81	.99
$> .2$	any	0	1	0	1

Ranges of posterior probabilities for Fieller' Set and Jeffreys' Set in two typical situations.

We will consider, as our functions of interest, the posterior probability of Fieller's set and the posterior probability of the Jeffreys' HPD credible set. Due to the special nature of Fieller's set, the only interesting cases to study are when the observed data are in $\mathcal{X}_1$ or in $\mathcal{X}_2$, because for $x \in \mathcal{X}_3$ Fieller's set is trivially robust!

Consider then, cases I and II in Table I, that is $x_1 = 2, x_2 = 2$, and $x_1 = 2, x_2 = .2$. Also set $n = 4$ and $\sigma = 1$. We report the posterior range of the probability of the 95% $C_F(x)$ and the posterior range of the Jeffreys' 95% credible set for various values of k and α. The numerical results show a little dependence ot the ranges on the value of α, at least for non extreme values (i.e. α close to 0 or close to 1). In case I, as k gets large, the posterior ranges of Fieller's set and Jeffreys' HPD set become huge. As soon as $k \geq \overline{y}$ (the point estimate of λ), the infimum of the posterior probability for the two sets is 0. Not surprisingly, the situation for case II is even worse. When $k = .1$, (that is, $k < \overline{y}$), the posterior ranges are quite small, with very high infima. As soon as $k \geq \overline{y}$, the ranges are inevitably $(0, 1)$. Note that, in case II, Fieller's set and Jeffreys' set are quite different: however, both of them suffer from a clear lack of robustness.

7. Conclusions

We have investigated several alternative ways of dealing with Fieller's problem. Apart from a subjective Bayesian analysis, no other approach seems to be satisfactory enough. Frequentist analysis cannot avoid the occurrence of unbounded (and unusable) confidence sets. We have also shown that Fieller's solution does not have good conditional properties. Its conditional behaviour is actually well illustrated by (29). Likelihood analysis of the problem fails to be operative: it is not clear how to use the profile likelihood or its modifications for inference about θ.

On the other hand, a Bayesian noninformative analysis does not have good frequentist coverage properties. A possible rejoinder to this criticism is that the frequentist behaviour of Bayesian procedures should be tested against some reasonable "conditional" distribution. However, formula (8) tells us that, even the conditional (on the ancillary statistic T) frequentist behaviour of Bayesian credible sets will not be satisfactory. In any case, a Bayesian noninformative analysis provides, at least, an operative posterior distribution for θ to work with. It is important to place matters in perspective: for these situation, like Fieller's problem, there simply are no methods of "objective" inference that will be viewed as broadly satisfactory. One could say that such problems are simply not amendable to any "objective" analysis; prior information is crucial and must be utilized. If, however, one is unwilling or unable to carry

out a true subjective Bayesian analysis, then a case can be made for having standard "default" analyses available. The Bayesian analyses with reference noninformative priors, such as that leading to (12), (14) or (16) are arguably the best candidates for such "default" analyses.

APPENDIX

Proof of Theorem 2. Consider first the case $x \in \mathcal{X}_1$. Let us define the matrix transformation (Lee, 1991) $V = BX$, where

$$B = \begin{pmatrix} \dfrac{n}{\sigma^2\sqrt{1+\theta^2}} & \dfrac{-n\theta}{\sigma^2\sqrt{1+\theta^2}} \\ 0 & \dfrac{n}{\sigma^2} \end{pmatrix}.$$

Also define $\delta = \omega_2 n/\sigma^2$. Then

$$V \sim N\left(\begin{pmatrix} 0 \\ \delta \end{pmatrix}, \Sigma\right), \qquad \Sigma = \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}, \qquad \rho = \frac{\theta}{\sqrt{1+\theta^2}}.$$

Then

$$P_\omega(C_F|\mathcal{X}_1) = P_\omega(V_1^2 \le z_{\alpha/2}^2 | V_2^2 \ge z_{\alpha/2}^2)$$

$$= \frac{\gamma - P_\omega(V_1^2 \le z_{\alpha/2}^2 | V_2^2 < z_{\alpha/2}^2)}{1 - \Phi(z_{\alpha/2} - \delta) + \Phi(-z_{\alpha/2} - \delta)}.$$

The quantity $h(\rho, \delta) = P_\omega(V_1^2 \le z_{\alpha/2}^2 | V_2^2 < z_{\alpha/2}^2)$ is such that $|h(\rho, 1) \le 1$. We can then interchange limit and integral signs to see that

$$\lim_{\rho \to 1} h(\rho, 0) = \gamma,$$

which concludes the proof. When $x \in \mathcal{X}_2$, calculations show that

$$P_\omega(C_F|\mathcal{X}_2) = \frac{\gamma - P_\omega(C_F \cap \mathcal{X}_1) - P_\omega(C_F \cap \mathcal{X}_3)}{1 - P_\omega(\mathcal{X}_1) - P_\omega(\mathcal{X}_3)}$$

$$= \frac{h(\rho, \delta) - P_\omega(\mathcal{X}_3)}{\Phi(z_{\alpha/2} - \delta) - \Phi(-z_{\alpha/2} - \delta) - P_\omega(\mathcal{X}_3)}$$

(since $C_F \bigcup \mathcal{X}_3 = \mathcal{X}_3$, for any parameter value, when we interpret C_F as a subset of $\mathcal{X}$). The above quantity is numerically minimised at $(\rho, \delta) = (0, 0)$. Therefore, for all $x \in \mathcal{X}_2$,

$$\gamma_c(x) = \frac{\int_{-z_{\alpha/2}}^{z_{\alpha/2}} \frac{1}{\sqrt{2\pi}} e^{-v_1^2/2} \left[\Phi(z_{\alpha/2}) - \Phi(-z_{\alpha/2})\right] dv_1 - P_0(\mathcal{X}_3)}{\gamma - P_0(\mathcal{X}_3)}$$

$$= \frac{(1-\alpha)^2 - (1 - e^{-z_{\alpha/2}^2/2})}{(1-\alpha) - (1 - e^{-z_{\alpha/2}^2/2})},$$

where $P_0(\mathcal{X}_3)$, the probability of $\mathcal{X}_3$ evaluated at $(\rho, \delta) = (0, 0)$, is from (3) the cdf of a central chi-square distribution with two degrees of freedom, evaluated at $z_{\alpha/2}^2$.

Proof of Theorem 3. For each value of ω, $C_F(x) \cup \mathcal{Y}_2 = \mathcal{Y}_2$, when we interpret C_F as a subset of the sample space. Then

$$P_\omega(C_F(X)|\mathcal{Y}_1) = \frac{P_\omega(C_F(X)) - P_\omega(\mathcal{Y}_2)}{1 - P_\omega(\mathcal{Y}_2)}$$

$$= \frac{1 - \alpha - \chi_2^2(n\|\omega\|/\sigma^2, z_{\alpha/2}^2)}{1 - \chi_2^2(n\|\omega\|/\sigma^2, z_{\alpha/2}^2)}.$$

The last quantity is decreasing in $\|\omega\|$, then for all $x \in \mathcal{Y}_1$,

$$\gamma_c'(x) = \frac{e^{-z_{\alpha/2}^2/2} - \alpha}{e^{-z_{\alpha/2}^2/2}}.$$

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